

NANGRONG, Thailand

WMO: 484360

Lat: 14.631N

Lon: 102.721E

Elev: 596

StdP: 14.38

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	59.0	61.4	50.7	56.0	67.5	53.5	62.1	67.7	14.3	76.8	12.3	75.1	1.3	10	0.313

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	19.6	100.5	77.6	98.4	77.8	96.6	77.9	81.4	92.0	80.8	91.3	80.3	90.5	3.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	154.3	84.6	78.4	150.7	84.1	78.0	148.9	83.8	45.6	92.3	45.0	91.5	44.5	90.9	93.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.8	10.3	8.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			54.6	103.2	2.3	2.1	52.9	104.8	51.6	106.0	50.3	107.2	48.6	108.8		
			52.2	82.9	2.3	0.9	50.5	83.5	49.2	84.1	47.9	84.6	46.3	85.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.2	76.2	80.4	84.5	86.7	85.6	85.3	84.2	83.8	82.6	81.3	79.0	76.1
	DBStd	4.76	4.59	4.72	4.25	3.26	2.58	1.95	2.13	1.86	1.94	2.14	3.67	4.50
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11738	813	852	1070	1100	1105	1058	1061	1048	979	971	871	810
	CDD65	6264	348	432	605	650	640	608	596	583	529	506	421	346
	CDH74	67235	3106	4696	7358	8103	7436	7124	6635	6275	5197	4709	3699	2897
	CDH80	30029	1295	2370	3983	4365	3476	3224	2823	2524	1863	1618	1393	1095
Wind	WSAvg	2.3	2.3	1.9	2.0	1.8	1.9	2.3	2.5	2.5	1.9	2.4	3.1	3.1
Precipitation	PrecAvg	46.5	0.2	0.7	1.9	3.0	5.9	5.3	5.4	7.8	9.6	5.4	1.2	0.2
	PrecMax	56.7	1.1	2.8	5.9	6.1	8.8	10.9	9.5	14.0	13.7	11.5	4.4	0.8
	PrecMin	36.2	0.0	0.0	0.1	0.6	2.5	2.0	2.3	3.6	4.6	3.1	0.0	0.0
	PrecStd	6.7	0.3	0.9	1.6	1.5	1.7	2.7	2.0	2.2	2.5	2.1	1.3	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.7	99.4	102.6	104.3	100.5	97.8	96.8	95.1	93.8	92.9	94.4	93.7
		MCWB	74.1	75.4	75.7	78.7	80.3	79.2	79.0	79.6	79.9	78.8	77.6	75.4
	2%	DB	92.2	97.0	100.3	101.1	97.5	95.7	94.4	93.3	91.9	90.7	91.6	91.1
		MCWB	74.1	75.3	76.2	78.6	80.0	79.3	79.0	79.2	79.8	78.8	77.1	75.3
	5%	DB	90.0	94.8	98.1	98.6	95.3	93.8	92.6	91.7	90.4	89.1	89.6	88.9
		MCWB	73.2	75.2	76.6	78.6	79.8	79.2	78.9	79.0	79.5	78.3	76.7	74.1
	10%	DB	87.4	92.3	95.7	96.1	93.1	92.0	90.9	90.0	88.7	87.7	87.5	86.3
		MCWB	72.4	74.9	76.6	78.3	79.8	79.1	78.7	78.7	79.0	77.9	75.9	73.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.1	79.5	80.2	81.6	82.3	81.5	81.1	81.1	81.6	80.8	80.3	78.3
		MCDB	89.0	92.8	94.2	96.6	93.6	91.9	91.1	90.8	90.0	88.4	89.0	88.7
	2%	WB	76.5	78.1	79.1	80.5	81.4	80.7	80.1	80.5	80.7	80.0	79.2	76.9
		MCDB	87.4	90.8	92.8	94.7	92.4	91.1	89.8	90.0	89.3	87.5	87.6	87.1
	5%	WB	75.2	77.1	78.3	79.8	80.8	80.1	79.7	79.9	80.0	79.3	78.2	75.6
		MCDB	85.1	89.6	91.6	93.4	91.6	90.0	89.2	88.8	88.0	86.4	85.9	85.2
	10%	WB	73.9	76.0	77.6	79.2	80.2	79.6	79.1	79.3	79.4	78.7	77.3	74.5
		MCDB	83.6	87.9	90.7	92.1	90.2	89.1	88.3	87.6	86.8	85.4	84.2	83.4
Mean Daily Temperature Range	5% DB	MDBR	22.4	23.2	21.4	19.6	15.9	14.8	14.3	13.8	12.8	13.0	16.4	19.9
		MCDBR	24.3	25.3	24.8	22.7	18.8	16.9	16.5	15.7	15.0	14.7	18.6	21.4
		MCWBR	7.7	6.8	5.3	5.0	4.7	4.2	4.5	4.6	4.9	4.7	5.9	6.9
	5% WB	MCDBR	20.3	21.8	20.9	20.2	16.5	15.3	14.9	14.7	13.9	13.4	15.8	19.1
	MCWBR	6.8	6.5	5.4	5.4	5.1	4.5	4.8	4.9	5.1	4.9	5.4	6.9	
Clear-Sky Solar Irradiance	taub	0.502	0.611	0.650	0.589	0.496	0.466	0.457	0.460	0.491	0.526	0.480	0.468	
	taud	2.026	1.774	1.726	1.908	2.212	2.300	2.320	2.297	2.200	2.068	2.178	2.159	
	Ebn at Noon	251	229	224	238	257	262	266	267	260	248	256	257	
	Edh at Noon	53	71	76	64	46	42	41	43	47	52	45	46	
All-Sky Solar Radiation	RadAvg	1580	1659	1695	1803	1758	1668	1581	1537	1474	1515	1547	1535	
	RadStd	95	69	73	81	101	64	93	89	83	112	88	54	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	+0.56	N/A	N/A	N/A	+0.47	+0.40	N/A	N/A	+236	+236

Nomenclature: See separate page